

GLOBAL REGULARITY FOR THE UNIDIRECTIONAL EULER-ALIGNMENT SYSTEM WITH SUPERCRITICAL DISSIPATION

CHANGHUI TAN AND LIUTANG XUE

ABSTRACT. We prove global well-posedness of the periodic unidirectional Euler–alignment system in every dimension for supercritical dissipation of order $0 < \alpha < 1$ and arbitrary non-vacuum initial data in the Sobolev class of the local theory. The key idea is to propagate simultaneously two moduli of continuity at critical scales for the density ρ and the potential gradient $\Gamma = \nabla \Lambda^{-\alpha} u$, both of which have scaling-invariant amplitudes. We also establish exponential alignment of the velocity and exponential convergence of the density to a traveling profile.

1. INTRODUCTION

The Euler–alignment system is a macroscopic model of collective motion in which individuals adjust their velocities toward those of their neighbors. It arises from the Cucker–Smale particle dynamics [8]; see also [12, 4] for hydrodynamic derivations and related models. The system takes the form

$$\begin{cases} \partial_t \rho + \nabla \cdot (\rho \mathbf{u}) = 0, \\ \partial_t \mathbf{u} + \mathbf{u} \cdot \nabla \mathbf{u} = \int_{\mathbb{R}^d} \psi(x - y) (\mathbf{u}(y) - \mathbf{u}(x)) \rho(y) dy, \end{cases} \quad (1.1)$$

where ρ is the density, $\mathbf{u} = (u_1, \dots, u_d)$ is the velocity, and the nonnegative communication kernel ψ describes the strength of the alignment interaction.

The regularity of solutions to (1.1) depends strongly on the singularity of ψ . For bounded Lipschitz kernels, alignment acts primarily as a nonlocal damping mechanism, and global regularity is governed by critical-threshold conditions on the initial data; see, for example, [22, 5]. Weakly singular but integrable kernels exhibit related threshold behavior [24].

In this paper, we consider strongly singular kernels of the form

$$\psi(x) = c_{d,\alpha} |x|^{-d-\alpha}, \quad x \in \mathbb{R}^d \setminus \{0\}, \quad 0 < \alpha < 2,$$

where $c_{d,\alpha} > 0$ is normalized so that the alignment force becomes

$$-\Lambda^\alpha(\rho \mathbf{u}) + \mathbf{u} \Lambda^\alpha \rho, \quad \Lambda^\alpha = (-\Delta)^{\alpha/2}.$$

Singular integrals are understood in the principal-value sense when necessary. The alignment force thus generates a density-weighted nonlocal diffusion, whose leading-order dissipative effect remains nondegenerate as long as the density stays strictly positive. In contrast, the presence of vacuum may lead to singularity formation even in one dimension [23, 1]. Motivated by this distinction, throughout this paper we consider (1.1) on the periodic domain $\mathbb{T}^d$ and assume that the initial density is bounded away from vacuum $\rho_0(x) > 0$. For periodic solutions, the singular integral operators above are understood as acting on periodic extensions.

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The one-dimensional theory for the singular Euler–alignment system has been established in [11, 20, 21]. A key feature is an additional structure that is not apparent from the velocity equation alone. The auxiliary quantity

$$G = \partial_x u - \Lambda^\alpha \rho$$

satisfies the continuity equation

$$\partial_t G + \partial_x(uG) = 0.$$

This remarkable conservation law provides an essential structural control on the coupling between the velocity and the density.

The dissipative mechanism is particularly transparent in the special class $G_0 = 0$. In this case $G \equiv 0$, and the system reduces to

$$\partial_t \rho + u \partial_x \rho = -\rho \Lambda^\alpha \rho, \quad u = -\partial_x \Lambda^{\alpha-2} \rho.$$

Thus the alignment interaction appears directly as a density-weighted fractional dissipation, closely related to fractional porous-medium flow with nonlocal pressure [3]. More generally, the conservation law for G , together with this dissipative mechanism, is a fundamental ingredient in the global well-posedness theory for arbitrary smooth non-vacuum initial data.

The theory is much less developed in multiple dimensions. For a general vector field $\mathbf{u}$, the natural analogue

$$G = \nabla \cdot \mathbf{u} - \Lambda^\alpha \rho$$

obeys

$$\partial_t G + \nabla \cdot (G\mathbf{u}) = (\nabla \cdot \mathbf{u})^2 - \text{tr}((\nabla \mathbf{u})^2).$$

The source term on the right is known as the spectral gap. It has no definite sign and does not vanish for a general multidimensional flow. Thus the conservation law for G , which is decisive in one dimension, is lost.

The absence of such a conserved quantity presents a fundamental difficulty in controlling the coupling between the density and the velocity gradient. In particular, the one-dimensional mechanism that leads to global regularity for arbitrary smooth non-vacuum data does not extend directly to general multidimensional flows. Consequently, the existing global well-posedness theory in multiple dimensions is largely perturbative. Global solutions have been constructed for small perturbations of aligned flocks or constant states; see, for example, [19, 9]. A global theory for arbitrary smooth non-vacuum initial data remains largely open.

Unidirectional flows form an important intermediate class between the one-dimensional and general multidimensional settings. Lear and Shvydkoy [16] considered velocities of the form

$$\mathbf{u}(x, t) = u(x, t)\mathbf{d}, \quad \mathbf{d} \in \mathbb{S}^{d-1},$$

an ansatz that is preserved by the Euler–alignment evolution. Without loss of generality, we take the distinguished direction to be $\mathbf{d} = e_1$. Then

$$\mathbf{u} = (u, 0, \dots, 0), \quad \nabla \cdot \mathbf{u} = \partial_1 u.$$

For such flows, the spectral gap vanishes identically: $(\nabla \cdot \mathbf{u})^2 - \text{tr}((\nabla \mathbf{u})^2) = 0$. Hence the auxiliary quantity

$$G = \partial_1 u - \Lambda^\alpha \rho$$

satisfies the conservation law

$$\partial_t G + \partial_1(uG) = 0.$$

Thus the key one-dimensional conservation structure is recovered. The unidirectional Euler–alignment system therefore takes the form

$$\begin{cases} \partial_t \rho + \partial_1(\rho u) = 0, \\ \partial_t u + u \partial_1 u = -\Lambda^\alpha(\rho u) + u \Lambda^\alpha \rho, \end{cases} \quad (1.2)$$

on $\mathbb{T}^d \times [0, T)$. Although the transport is directed along x_1 , both ρ and u depend on all d spatial variables, and Λ^α acts in every direction. Accordingly, this is not merely a one-dimensional problem embedded in a higher-dimensional space. In particular, the relation

$$\partial_1 u = \Lambda^\alpha \rho + G \quad (1.3)$$

controls only the derivative of u in the transport direction and provides no direct control of the transverse derivatives $\partial_k u$, $k \neq 1$. Even in the special class $G_0 = 0$, where $G \equiv 0$, this relation alone does not close the transverse regularity estimates. This constitutes the main structural distinction between the unidirectional system and its genuinely one-dimensional counterpart.

The fractional Burgers equation is a second useful benchmark. If the density ρ in the velocity equation is frozen at one, the alignment operator becomes $-\Lambda^\alpha u$, leading formally to

$$\partial_t u + u \partial_1 u = -\Lambda^\alpha u.$$

Solutions of fractional Burgers remain globally regular for $\alpha \geq 1$, while finite-time singularities may occur for $0 < \alpha < 1$ [13]. In the one-dimensional Euler–alignment system, the variable density weight enhances the dissipative mechanism sufficiently to prevent this Burgers-type breakdown for non-vacuum data. A central question left by the preceding theory was whether this enhancement survives in the multidimensional unidirectional setting, or whether it acts effectively only in the distinguished direction x_1 .

Lear and Shvydkoy [16] established global regularity for the unidirectional system in the subcritical range $1 < \alpha < 2$. The terminology is consistent with the natural scaling

$$\rho_\lambda(x, t) = \rho(\lambda x, \lambda^\alpha t), \quad u_\lambda(x, t) = \lambda^{\alpha-1} u(\lambda x, \lambda^\alpha t). \quad (1.4)$$

When $\alpha > 1$, the velocity amplitude is subcritical under this scaling. In particular, the fractional dissipation is strong enough to dominate the Burgers-type transport, much as in the subcritical fractional Burgers equation. In this regime, one may therefore treat the velocity equation directly as a dissipative transport equation, without relying on the more delicate one-dimensional mechanism. Once u is controlled, the density can be recovered from

$$\rho = \partial_1 \Lambda^{-\alpha} u - \Lambda^{-\alpha} G.$$

Indeed, for $\alpha > 1$, the operator $\partial_1 \Lambda^{-\alpha}$ has negative order $1 - \alpha < 0$ and is therefore smoothing. Together with the control of the conserved quantity G , this allows the regularity of ρ to be recovered from that of u .

The critical case $\alpha = 1$ is considerably more subtle. Lear [15] obtained global regularity under the special null-entropy condition $G_0 = 0$, as well as a separate small-data result for general initial entropy. The first large-data global well-posedness result for general unidirectional flows at the critical exponent was obtained by Li et al. [18]. In the critical regime, both unknowns have scaling-invariant amplitudes,

$$\rho_\lambda(x, t) = \rho(\lambda x, \lambda t), \quad u_\lambda(x, t) = u(\lambda x, \lambda t),$$

and the reconstruction of the density becomes borderline:

$$\rho = \partial_1 \Lambda^{-1} u - \Lambda^{-1} G.$$

The leading operator $\partial_1 \Lambda^{-1}$ is a Riesz transform and is not bounded from L^∞ to L^∞ . Thus control of the velocity alone no longer yields sufficient control of the density. This difficulty also appears directly in the alignment force: the commutator remainder contains simultaneous increments of ρ and u and cannot be controlled by a modulus of continuity for u alone.

The main idea of [18] is therefore to propagate moduli of continuity for ρ and u simultaneously. Since both quantities have critical, scaling-invariant amplitudes at $\alpha = 1$, the two moduli can be chosen compatibly so that their breakthrough scenarios are coupled and ruled out together. This simultaneous propagation of multiple moduli of continuity is a genuinely system-level extension

of the modulus method developed for critical scalar equations such as the critical Burgers and quasi-geostrophic equations [13, 14].

The supercritical regime $0 < \alpha < 1$ presents a fundamentally different difficulty. Here the fractional dissipation is too weak, by itself, to control Burgers-type steepening. Thus any large-data global regularity mechanism must exploit the special structure inherited from the one-dimensional Euler–alignment system. However, in the multidimensional unidirectional setting, the relation (1.3) controls only the derivative in the distinguished direction x_1 . It provides no direct control of the transverse derivatives of u , while in the supercritical range these transverse derivatives cannot be handled by dissipation alone.

In [18], this difficulty led to the continuation criterion

$$u \in L_t^\infty C_x^\sigma, \quad \text{for some } \sigma \in (1 - \alpha, 1). \quad (1.5)$$

In view of the scaling (1.4), the exponent $1 - \alpha$ is critical for the velocity. Thus, once a Hölder bound strictly above this critical level is available, the remaining nonlinear effects become perturbative and the solution can be continued. The essential unresolved issue, however, is to prove such a bound dynamically from arbitrary large initial data; this was not addressed in [18].

The goal of the present work is to resolve precisely this issue. We prove global well-posedness in the full supercritical range without imposing any additional continuation criterion or a priori Hölder control on the velocity. In particular, the transverse regularity is obtained dynamically, despite the weakness of the dissipation and the fact that the structural relation acts directly only in the distinguished direction. We also prove exponential alignment of the velocity and convergence of the density to a traveling profile.

Our main result is stated as follows.

Theorem 1.1 (Global classical solutions and asymptotic alignment). *Let $d \geq 1$, $0 < \alpha < 1$, and let $m \in \mathbb{N}$ satisfy $m > d/2 + 1$. Assume that*

$$(\rho_0, u_0) \in H^{m+\alpha}(\mathbb{T}^d) \times H^{m+1}(\mathbb{T}^d), \quad \inf_{x \in \mathbb{T}^d} \rho_0(x) > 0. \quad (1.6)$$

Then the unidirectional Euler–alignment system (1.2) admits a unique global classical solution satisfying

$$\begin{aligned} \rho &\in C_w([0, \infty); H^{m+\alpha}(\mathbb{T}^d)), \\ u &\in C_w([0, \infty); H^{m+1}(\mathbb{T}^d)) \cap L_{\text{loc}}^2([0, \infty); \dot{H}^{m+1+\alpha/2}(\mathbb{T}^d)). \end{aligned} \quad (1.7)$$

Here C_w denotes weak continuity in time.

Moreover, there exist a positive profile $\rho_\infty \in W^{1,\infty}(\mathbb{T}^d)$ with the same total mass as ρ_0 and constants $C < \infty$ and $\kappa > 0$, depending only on the initial data and the fixed parameters, such that

$$\|\rho(\cdot, t) - \rho_\infty(\cdot - \bar{u}te_1)\|_{L^\infty(\mathbb{T}^d)} \leq Ce^{-\kappa t}, \quad (1.8)$$

$$\|u(\cdot, t) - \bar{u}\|_{W^{1,\infty}(\mathbb{T}^d)} \leq Ce^{-\kappa t}, \quad t \geq 0, \quad (1.9)$$

where the density-weighted average velocity is

$$\bar{u} := \frac{\int_{\mathbb{T}^d} \rho_0(x) u_0(x) \, dx}{\int_{\mathbb{T}^d} \rho_0(x) \, dx}.$$

Let us describe the main idea and strategy of the proof. The key is to introduce the potential gradient

$$\Gamma := \nabla \Lambda^{-\alpha} u,$$

where $\Lambda^{-\alpha}$ annihilates the zero Fourier mode. There are two important reasons for this choice. First, Γ has scaling-invariant amplitude. Indeed, under the scaling (1.4),

$$\rho_\lambda(x, t) = \rho(\lambda x, \lambda^\alpha t), \quad \Gamma_\lambda(x, t) = \Gamma(\lambda x, \lambda^\alpha t).$$

Thus ρ and Γ live at the same critical scale, making them natural quantities for a simultaneous modulus-of-continuity argument. Second, Γ retains the crucial Euler–alignment structure encoded in (1.3). Applying $\nabla \Lambda^{-\alpha}$ to the velocity equation and using

$$\partial_1 u = \Lambda^\alpha \rho + G,$$

one finds a cancellation of all derivatives of the density and obtains the transport-diffusion equation

$$\partial_t \Gamma + u \partial_1 \Gamma + \rho \Lambda^\alpha \Gamma = f, \quad f := -[\nabla \Lambda^{-\alpha}, u] G. \quad (1.10)$$

This cancellation is essential: the leading part of the Γ -equation is a density-weighted fractional diffusion, while the remaining forcing has the commutator structure above.

Motivated by the effectiveness of simultaneously propagating moduli of continuity in the critical case [18], we seek to propagate compatible moduli for ρ and Γ . The main new difficulty is the commutator forcing f . Its control is carried out in two steps.

We first establish an L^∞ bound for Γ , which is needed to initiate the modulus argument. Since the commutator prevents a direct application of the maximum principle, we begin with the endpoint commutator estimate

$$\|f\|_{L^p} \leq C_{d,\alpha,p} \|G\|_{L^\infty} \|\Gamma\|_{L^p}, \quad 1 < p < \infty.$$

Together with the uniform bound on G and the Córdoba–Córdoba inequality [7], this closes the finite- p estimate without requiring any control of density gradients. Since the commutator estimate does not extend directly to $p = \infty$, we derive an $L^p \rightarrow L^\infty$ smoothing estimate for the corresponding homogeneous equation by fractional Moser iteration. Duhamel’s formula then yields an L^∞ bound for Γ . Combining velocity alignment with a nonlinear dissipation estimate further gives exponential decay of $\|\Gamma(t)\|_{L^\infty}$.

We then propagate a fixed modulus for ρ and an exponentially decreasing modulus for Γ ,

$$\omega_1(\xi) = \omega_\lambda^\delta(\xi), \quad \omega_2(\xi, t) = e^{-\kappa t} \omega_\lambda^\delta(\xi),$$

where ω_λ^δ is the family defined in (4.3). The common critical scaling of ρ and Γ is crucial here: it allows us first to choose the amplitude δ sufficiently small so that the transport and variable-coefficient contributions can be absorbed by the dissipation, and then to choose the spatial scale λ sufficiently small to accommodate arbitrary initial data. The main work in the Γ breakthrough scenario is to estimate the difference

$$f(x, t) - f(y, t)$$

at a possible contact pair. By exploiting both the commutator structure and the simultaneous moduli for ρ and Γ , we show that this difference has a modulus compatible with the dissipative term. This allows the dissipation to rule out both the density and Γ breakthrough scenarios.

The propagated moduli yield

$$\sup_{t \geq 0} \|\nabla \rho(t)\|_{L^\infty} < \infty, \quad [\Gamma(t)]_{\text{Lip}} \lesssim e^{-\kappa t},$$

and, through the reconstruction of u from Γ , also

$$\|\nabla u(t)\|_{L^\infty} \lesssim e^{-\kappa t}.$$

This is substantially stronger than the conditional Hölder control required in (1.5): rather than assuming that u remains in C^σ for some $\sigma > 1 - \alpha$, the argument dynamically produces a global Lipschitz bound for the velocity. These estimates close the continuation criterion and yield global regularity.

The rest of the paper is organized as follows. Section 2 collects the local well-posedness theory and continuation criterion, together with the basic a priori bounds and the endpoint commutator estimate. Section 3 derives the evolution equation for the potential gradient Γ and establishes its L^p and L^∞ bounds, followed by exponential decay. Section 4 develops the simultaneous modulus-of-continuity argument for ρ and Γ , including the breakthrough analysis and the key estimate on the commutator

forcing. [Section 5](#) closes the continuation argument and proves exponential alignment of the velocity and convergence of the density to a traveling profile, completing the proof of [Theorem 1.1](#). Finally, [Section A](#) provides a proof of the periodic endpoint commutator estimate.

Notations. We work on $\mathbb{T}^d = (\mathbb{R}/2\pi\mathbb{Z})^d$ with its geodesic distance $d_{\mathbb{T}^d}$. Our Fourier convention is $\widehat{f}(\xi) = (2\pi)^{-d} \int_{\mathbb{T}^d} f(x) e^{-ix \cdot \xi} dx$, $\xi \in \mathbb{Z}^d$, and $\langle f \rangle := \widehat{f}(0)$. For $s \neq 0$, the operator $\Lambda^s = (-\Delta)^{s/2}$ is defined by

$$\widehat{\Lambda^s f}(\xi) = \begin{cases} |\xi|^s \widehat{f}(\xi), & \xi \neq 0, \\ 0, & \xi = 0. \end{cases} \quad (1.11)$$

In particular, negative powers and the Riesz transforms $R_k := \partial_k \Lambda^{-1}$ annihilate constants.

Identifying f with its periodic extension to $\mathbb{R}^d$, we write

$$\Lambda^\alpha f(x) = c_{d,\alpha} \text{p.v.} \int_{\mathbb{R}^d} \frac{f(x) - f(y)}{|x - y|^{d+\alpha}} dy, \quad 0 < \alpha < 2, \quad (1.12)$$

where $c_{d,\alpha} > 0$ is normalized so that Λ^α has Fourier symbol $|\xi|^\alpha$. For $0 < \alpha < 1$ and $f \in C^1(\mathbb{T}^d)$, the integral is absolutely convergent. All integrals over $\mathbb{R}^d$ involving periodic functions are understood in this sense. The phrase *fixed periodic normalization* refers to these choices of torus, Fourier transform, and kernel. Constants arising from this normalization depend only on d, α and may change from line to line.

2. PRELIMINARIES

2.1. Local well-posedness theory and a priori bounds. We collect the known results used below, following [\[18, Section 2\]](#), and state them without proof.

First, we state the local well-posedness theory and regularity criterion for smooth solutions of the system [\(1.2\)](#).

Proposition 2.1. *Consider the system [\(1.2\)](#) with initial data satisfying [\(1.6\)](#). Then there exist a maximal time $T_* \in (0, \infty]$ and a unique non-vacuum classical solution (ρ, u) on $[0, T_*)$ such that*

$$\begin{aligned} \rho &\in C_w([0, T_*); H^{m+\alpha}(\mathbb{T}^d)), \\ u &\in C_w([0, T_*); H^{m+1}(\mathbb{T}^d)) \cap L_{\text{loc}}^2([0, T_*); \dot{H}^{m+1+\alpha/2}(\mathbb{T}^d)). \end{aligned} \quad (2.1)$$

Here C_w denotes weak continuity in time, and T_* is the maximal existence time in the solution class [\(2.1\)](#). Moreover, if $T_* < \infty$, then

$$\sup_{0 \leq t < T_*} \left(\|\nabla \rho(\cdot, t)\|_{L^\infty(\mathbb{T}^d)} + \|\nabla u(\cdot, t)\|_{L^\infty(\mathbb{T}^d)} \right) = \infty. \quad (2.2)$$

The local theory and continuation criterion are given in [\[16, Theorem 1.1\]](#).

Next, we state some a priori bounds. Denote the auxiliary quantity

$$G := \partial_1 u - \Lambda^\alpha \rho.$$

It satisfies the following conservation law:

$$\partial_t G + \partial_1(Gu) = 0.$$

Define $F := G/\rho$. Then F satisfies the transport equation

$$(\partial_t + u\partial_1)F = 0. \quad (2.3)$$

Consequently, we have

$$\|F(\cdot, t)\|_{L^\infty} = \|F_0\|_{L^\infty},$$

where $F_0 := F(\cdot, 0)$.

We have the following a priori bounds on the density: there exist constants $0 < \underline{\rho} \leq \bar{\rho} < \infty$, depending only on the initial data, d, α , and the periodic normalization, such that

$$\underline{\rho} \leq \rho(x, t) \leq \bar{\rho}, \quad (x, t) \in \mathbb{T}^d \times [0, T_*]. \quad (2.4)$$

See, e.g., [18, Proposition 2.2]. This then implies

$$\|G(\cdot, t)\|_{L^\infty} = \|\rho(\cdot, t)F(\cdot, t)\|_{L^\infty} \leq \bar{\rho}\|F_0\|_{L^\infty}. \quad (2.5)$$

Finally, the velocity satisfies the maximum principle $\|u(t)\|_{L^\infty} \leq \|u_0\|_{L^\infty}$ and the following flocking estimate; see [18, Lemma 2.3] and [22]. Define

$$V(t) := \sup_{x, y \in \mathbb{T}^d} |u(x, t) - u(y, t)|, \quad V_0 := V(0).$$

Then there exists $\kappa > 0$, depending only on d, α and the initial mass, such that

$$V(t) \leq V_0 e^{-\kappa t}, \quad 0 \leq t < T_*. \quad (2.6)$$

2.2. The endpoint commutator. For $k \in \{1, \dots, d\}$, write

$$\mathcal{R}_{k, \alpha} := \partial_k \Lambda^{-\alpha}.$$

We shall use the following Kato–Ponce type commutator estimate to control the forcing term f in (1.10).

Lemma 2.2. *Let $0 < \alpha < 1$ and $1 < p < \infty$. For every $k \in \{1, \dots, d\}$ and every $v, h \in C^1(\mathbb{T}^d)$,*

$$\|[\mathcal{R}_{k, \alpha}, v]h\|_{L^p(\mathbb{T}^d)} \leq C_{d, \alpha, p} \|\nabla \Lambda^{-\alpha} v\|_{L^p(\mathbb{T}^d)} \|h\|_{L^\infty(\mathbb{T}^d)}. \quad (2.7)$$

The constant also depends on the fixed periodic normalization.

The estimate (2.7) is an endpoint commutator estimate in the sense that the second factor is measured in L^∞ . Its Euclidean counterpart on $\mathbb{R}^d$ follows, for example, from [17, Corollary 1.4(2)], with $s = s_1 = 1 - \alpha$ and $s_2 = 0$. The corresponding estimate on the periodic domain $\mathbb{T}^d$ can be obtained by a standard transference/localization argument; see also [2] for periodic fractional Leibniz and related commutator estimates. For completeness, we provide a proof in Section A.

3. A PRIORI BOUNDS FOR THE POTENTIAL GRADIENT

We introduce the potential ϕ and its gradient Γ :

$$\phi := \Lambda^{-\alpha}(u - \langle u \rangle), \quad \text{and} \quad \Gamma := \nabla \phi = \nabla \Lambda^{-\alpha}(u - \langle u \rangle).$$

On periodic lifts, with the period rescaled, Γ satisfies the same critical scaling as ρ :

$$\Gamma_\lambda(x, t) = \Gamma(\lambda x, \lambda^\alpha t).$$

In this section, we establish the L^∞ bound for Γ . The estimate will be used in the modulus-of-continuity argument in the next section.

Throughout this section, we use the structural relation

$$\partial_1 u = \Lambda^\alpha \rho + G, \quad (3.1)$$

together with the a priori bounds obtained in Section 2,

$$0 < \underline{\rho} \leq \rho(x, t) \leq \bar{\rho}, \quad \|G(t)\|_{L^\infty} \leq G_*, \quad 0 \leq t < T_*.$$

Here and below, constants may depend on the fixed periodic normalization. We take $G_* := \bar{\rho}\|F_0\|_{L^\infty}$, as provided by (2.5).

3.1. The potential-gradient equation. For $k \in \{1, \dots, d\}$, denote

$$\Gamma := (\Gamma_1, \dots, \Gamma_d), \quad \text{where} \quad \Gamma_k := \mathcal{R}_{k,\alpha} u, \quad \text{and} \quad \mathcal{R}_{k,\alpha} := \partial_k \Lambda^{-\alpha}.$$

We first derive the equation satisfied by Γ . By (3.1), the velocity equation can be rewritten as

$$\partial_t u = -u \partial_1 u - \Lambda^\alpha(\rho u) + u \Lambda^\alpha \rho = -\Lambda^\alpha(\rho u) - u G.$$

Applying $\mathcal{R}_{k,\alpha}$ and using $\mathcal{R}_{k,\alpha} \Lambda^\alpha = \partial_k$, we obtain

$$\partial_t \Gamma_k = -\partial_k(\rho u) - \mathcal{R}_{k,\alpha}(u G).$$

Moreover,

$$\Lambda^\alpha \Gamma_k = \partial_k u, \quad \text{and} \quad \partial_1 \Gamma_k = \partial_k \rho + \mathcal{R}_{k,\alpha} G.$$

Consequently,

$$\begin{aligned} \partial_t \Gamma_k + u \partial_1 \Gamma_k + \rho \Lambda^\alpha \Gamma_k &= -\partial_k(\rho u) - \mathcal{R}_{k,\alpha}(u G) + u(\partial_k \rho + \mathcal{R}_{k,\alpha} G) + \rho \partial_k u \\ &= u \mathcal{R}_{k,\alpha} G - \mathcal{R}_{k,\alpha}(u G) = -[\mathcal{R}_{k,\alpha}, u] G. \end{aligned}$$

Thus, Γ satisfies the equation

$$\partial_t \Gamma + u \partial_1 \Gamma + \rho \Lambda^\alpha \Gamma = f, \tag{3.2}$$

where the forcing term has a commutator form

$$f = (f_1, \dots, f_d), \quad f_k := -[\mathcal{R}_{k,\alpha}, u] G.$$

3.2. The L^p bound. We first obtain an L^p bound for Γ for every finite p .

Proposition 3.1. *Let $2 \leq p < \infty$. Then*

$$\|\Gamma(t)\|_{L^p} \leq e^{C_p G_* t} \|\Gamma_0\|_{L^p}, \quad 0 \leq t < T_*. \tag{3.3}$$

Proof. Multiplying (3.2) by $p|\Gamma|^{p-2}\Gamma$ and integrating over $\mathbb{T}^d$, we obtain

$$\begin{aligned} \frac{d}{dt} \int_{\mathbb{T}^d} |\Gamma|^p dx &= \int_{\mathbb{T}^d} (\partial_1 u) |\Gamma|^p dx - p \int_{\mathbb{T}^d} \rho |\Gamma|^{p-2} \Gamma \cdot \Lambda^\alpha \Gamma dx + p \int_{\mathbb{T}^d} f \cdot |\Gamma|^{p-2} \Gamma dx \\ &= \int_{\mathbb{T}^d} \Lambda^\alpha \rho |\Gamma|^p dx + \int_{\mathbb{T}^d} G |\Gamma|^p dx - p \int_{\mathbb{T}^d} \rho |\Gamma|^{p-2} \Gamma \cdot \Lambda^\alpha \Gamma dx + p \int_{\mathbb{T}^d} f \cdot |\Gamma|^{p-2} \Gamma dx, \end{aligned} \tag{3.4}$$

where we have used the relation (3.1).

Applying the vector-valued Córdoba–Córdoba inequality [7] to the convex function $s \mapsto |s|^p$ with $s = \Gamma$, we have

$$\Lambda^\alpha(|\Gamma|^p) \leq p |\Gamma|^{p-2} \Gamma \cdot \Lambda^\alpha \Gamma. \tag{3.5}$$

Therefore,

$$\int_{\mathbb{T}^d} \Lambda^\alpha \rho |\Gamma|^p dx = \int_{\mathbb{T}^d} \rho \Lambda^\alpha(|\Gamma|^p) dx \leq p \int_{\mathbb{T}^d} \rho |\Gamma|^{p-2} \Gamma \cdot \Lambda^\alpha \Gamma dx.$$

Noting that

$$\left\| |\Gamma|^{p-2} \Gamma \right\|_{L^{\frac{p}{p-1}}} = \left(\int_{\mathbb{T}^d} |\Gamma|^{(p-1) \cdot \frac{p}{p-1}} dx \right)^{\frac{p-1}{p}} = \|\Gamma\|_{L^p}^{p-1},$$

we apply Hölder's inequality to the remaining two terms in (3.4) and obtain

$$\frac{d}{dt} \|\Gamma\|_{L^p}^p \leq \|G\|_{L^\infty} \|\Gamma\|_{L^p}^p + p \|f\|_{L^p} \|\Gamma\|_{L^p}^{p-1}.$$

Finally, for the forcing term, we apply commutator estimate (2.7) and obtain

$$\|f_k\|_{L^p} = \|[\mathcal{R}_{k,\alpha}, u] G\|_{L^p} \leq C_{d,\alpha,p} \|\nabla \Lambda^{-\alpha} u\|_{L^p} \|G\|_{L^\infty} \leq C_{d,\alpha,p} G_* \|\Gamma\|_{L^p}. \tag{3.6}$$

Applying all the estimates to (3.4), we find

$$\frac{d}{dt} \|\Gamma\|_{L^p}^p \leq C_p G_* \|\Gamma\|_{L^p}^p.$$

Grönwall's inequality gives (3.3). □

As a direct consequence of (3.3) and (3.6), we have the bound

$$\|f(t)\|_{L^p} \leq C_p G_* e^{C_p G_* t} \|\Gamma_0\|_{L^p}. \quad (3.7)$$

3.3. The L^∞ bound. Proposition 3.1 does not directly extend to the case when $p = \infty$. In particular, the estimate in (3.6) ultimately uses the L^p -boundedness of the periodic Riesz transforms, while the Riesz transforms do not map L^∞ to L^∞ . Thus we do not have a direct estimate of the form

$$\|f\|_{L^\infty} \leq C G_* \|\Gamma\|_{L^\infty}.$$

To obtain an L^∞ bound, we first use the dissipation to obtain an $L^p \rightarrow L^\infty$ smoothing estimate for the homogeneous equation corresponding to (3.2):

$$\partial_t v + u \partial_1 v + \rho \Lambda^\alpha v = 0, \quad v(\cdot, s) = v_s, \quad (3.8)$$

where v is scalar-valued. The estimate will be applied componentwise to Γ .

We first state the following maximum principle.

Lemma 3.2. *For every scalar solution of (3.8) and every $s \leq t$,*

$$\|v(t)\|_{L^\infty} \leq \|v_s\|_{L^\infty}. \quad (3.9)$$

Proof. Suppose that v attains a positive spatial maximum at x_t . Then

$$\partial_1 v(x_t, t) = 0, \quad \Lambda^\alpha v(x_t, t) \geq 0.$$

Since $\rho(x_t, t) > 0$, (3.8) implies

$$\frac{d^+}{dt} \max_x v(x, t) \leq 0.$$

Applying the same argument to $-v$, we obtain (3.9). $\square$

Next, we derive the $L^p \rightarrow L^\infty$ smoothing estimate, using the Moser iteration.

Proposition 3.3. *Let $2 \leq p < \infty$. Then every smooth solution of (3.8) satisfies*

$$\|v(t)\|_{L^\infty} \leq C \left(1 + (t-s)^{-d/(\alpha p)}\right) \|v_s\|_{L^p}, \quad 0 \leq s < t < T_*. \quad (3.10)$$

Here C depends only on d, α, p, ρ, G_ , and the fixed periodic normalization.*

Proof. We first derive an L^q energy inequality which is uniform in q . Fix $q \geq 2$. The same calculation as (3.4) yields

$$\begin{aligned} \frac{d}{dt} \|v\|_{L^q}^q &= \int_{\mathbb{T}^d} \rho \left(\Lambda^\alpha (|v|^q) - q |v|^{q-2} v \Lambda^\alpha v \right) dx + \int_{\mathbb{T}^d} G |v|^q dx \\ &\leq \rho \int_{\mathbb{T}^d} \left(\Lambda^\alpha (|v|^q) - q |v|^{q-2} v \Lambda^\alpha v \right) dx + G_* \|v\|_{L^q}^q, \end{aligned}$$

where we have used the Córdoba–Córdoba inequality (3.5). Then we apply the periodic version of the Stroock–Varopoulos inequality [10, Lemma 5.1]:

$$\int_{\mathbb{T}^d} q |v|^{q-2} v \Lambda^\alpha v dx \geq \frac{4(q-1)}{q} \left\| \Lambda^{\alpha/2} (|v|^{q/2}) \right\|_{L^2}^2 \geq 2 \left\| \Lambda^{\alpha/2} (|v|^{q/2}) \right\|_{L^2}^2.$$

It yields

$$\frac{d}{dt} \|v\|_{L^q}^q + 2\rho \left\| \Lambda^{\alpha/2} (|v|^{q/2}) \right\|_{L^2}^2 \leq G_* \|v\|_{L^q}^q. \quad (3.11)$$

In particular, discarding the dissipative term gives

$$\|v(t)\|_{L^q}^q \leq e^{G_*(t-\tau)} \|v(\tau)\|_{L^q}^q, \quad s \leq \tau \leq t. \quad (3.12)$$

Next, we use the dissipation to increase the integrability exponent. Set

$$\chi = \frac{d}{d-\alpha} > 1.$$

The periodic fractional Sobolev inequality gives

$$\|h\|_{L^{2\chi}}^2 \leq C \left(\left\| \Lambda^{\alpha/2} h \right\|_{L^2}^2 + \|h\|_{L^2}^2 \right).$$

Applying this estimate to $h = |v|^{q/2}$, we obtain

$$\|v\|_{L^{q\chi}}^q \leq C \left\| \Lambda^{\alpha/2} (|v|^{q/2}) \right\|_{L^2}^2 + C \|v\|_{L^q}^q.$$

Combining this inequality with (3.11) yields

$$\frac{d}{dt} \|v\|_{L^q}^q + c \|v\|_{L^{q\chi}}^q \leq C \|v\|_{L^q}^q. \quad (3.13)$$

Let

$$\tau < \sigma, \quad 0 < \sigma - \tau \leq 1.$$

Integrating (3.13) on $[\tau, \sigma]$, we get

$$\int_{\tau}^{\sigma} \|v(r)\|_{L^{q\chi}}^q dr \leq C \|v(\tau)\|_{L^q}^q, \quad (3.14)$$

where C is independent of $q \geq 2$.

We now perform the Moser iteration. Fix $0 < t - s \leq 1$. Define

$$q_n = p\chi^n, \quad t_n = s + (t - s)(1 - 2^{-n}), \quad n = 0, 1, 2, \dots,$$

so that

$$t_0 = s, \quad t_n \uparrow t, \quad t_{n+1} - t_n = 2^{-n-1}(t - s).$$

Applying (3.14) with

$$q = q_n, \quad \tau = t_n, \quad \sigma = t_{n+1},$$

and using $q_{n+1} = q_n\chi$, we obtain

$$\int_{t_n}^{t_{n+1}} \|v(r)\|_{L^{q_{n+1}}}^{q_n} dr \leq C \|v(t_n)\|_{L^{q_n}}^{q_n}.$$

Hence there exists $r_n \in [t_n, t_{n+1}]$ such that

$$\|v(r_n)\|_{L^{q_{n+1}}} = \left(\frac{1}{t_{n+1} - t_n} \int_{t_n}^{t_{n+1}} \|v(r)\|_{L^{q_{n+1}}}^{q_n} dr \right)^{1/q_n} \leq \left(\frac{C2^{n+1}}{t - s} \right)^{1/q_n} \|v(t_n)\|_{L^{q_n}}.$$

Using (3.12) at exponent q_{n+1} , from r_n to t_{n+1} , gives

$$\|v(t_{n+1})\|_{L^{q_{n+1}}} \leq e^{G^*/q_{n+1}} \|v(r_n)\|_{L^{q_{n+1}}} \leq \left(\frac{C2^{n+1}}{t - s} \right)^{1/q_n} e^{G^*/q_{n+1}} \|v(t_n)\|_{L^{q_n}}. \quad (3.15)$$

Iterating (3.15) from $n = 0$ to $N - 1$, we find

$$\|v(t_N)\|_{L^{q_N}} \leq \prod_{n=0}^{N-1} \left(C^{1/q_n} 2^{(n+1)/q_n} e^{G^*/q_{n+1}} \right) \cdot (t - s)^{-\sum_{n=0}^{N-1} 1/q_n} \|v_s\|_{L^p}. \quad (3.16)$$

Since

$$\sum_{n=0}^{\infty} \frac{1}{q_n} = \frac{1}{p} \sum_{n=0}^{\infty} \chi^{-n} = \frac{d}{\alpha p}, \quad \text{and} \quad \sum_{n=0}^{\infty} \frac{n+1}{q_n} < \infty,$$

the product in (3.16) converges. Letting $N \rightarrow \infty$, we conclude that

$$\|v(t)\|_{L^\infty} \leq C(t - s)^{-d/(\alpha p)} \|v_s\|_{L^p}, \quad 0 < t - s \leq 1.$$

For $t > s + 1$, we apply the maximum principle (3.9) and get

$$\|v(t)\|_{L^\infty} \leq \|v(s+1)\|_{L^\infty} \leq C \|v_s\|_{L^p}.$$

This finishes the proof. $\square$

We are ready to obtain the L^∞ bound on Γ . We apply Duhamel's formula and use the smoothing estimate to control the forcing term.

Proposition 3.4. *For every $0 < T < T_*$, there exists $C_T > 0$, depending only on $T, d, \alpha, \underline{\rho}, G_*$, and the periodic normalization, such that*

$$\sup_{0 \leq t \leq T} \|\Gamma(t)\|_{L^\infty} \leq C_T \|\Gamma_0\|_{L^\infty}. \quad (3.17)$$

Proof. Denote $S(t, s)$ the solution operator of the homogeneous equation (3.8), namely

$$v(t) = S(t, s) v_s.$$

Apply Duhamel's formula for each component of (3.2):

$$\Gamma_k(t) = S(t, 0) \Gamma_{k,0} + \int_0^t S(t, s) f_k(s) ds. \quad (3.18)$$

For the first term, we simply apply the maximum principle (3.9) and get

$$\|S(t, 0) \Gamma_{k,0}\|_{L^\infty} \leq \|\Gamma_{k,0}\|_{L^\infty}.$$

For the second term, fix $p > \max\{2, \frac{d}{\alpha}\}$. The smoothing estimate (3.10) yields

$$\left\| \int_0^t S(t, s) f_k(s) ds \right\|_{L^\infty} \leq C \int_0^t \left(1 + (t-s)^{-\beta}\right) \|f_k(s)\|_{L^p} ds,$$

where $\beta := \frac{d}{\alpha p} \in (0, 1)$. Using (3.7), we further estimate

$$\left\| \int_0^t S(t, s) f_k(s) ds \right\|_{L^\infty} \leq C \|\Gamma_0\|_{L^p} \int_0^t \left(1 + (t-s)^{-\beta}\right) e^{Cs} ds \leq C \|\Gamma_0\|_{L^\infty} \left(T + \frac{T^{1-\beta}}{1-\beta}\right) e^{CT}.$$

Combining the two terms yields (3.17) with $C_T = C \left[1 + \left(T + \frac{T^{1-\beta}}{1-\beta}\right) e^{CT}\right]$. In particular, these constants remain bounded as $T \uparrow T_* < \infty$. $\square$

3.4. Exponential decay of the potential gradient. Using the flocking estimate in (2.6), we can upgrade the preceding bounds for Γ to exponential decay.

Proposition 3.5 (Decay of the potential gradient). *For every fixed $2 \leq p < \infty$, there exists $C_p > 0$, depending only on the initial data, d, α, p , and the fixed periodic normalization, such that*

$$\|\Gamma(t)\|_{L^p} \leq C_p e^{-\kappa t}, \quad 0 \leq t < T_*. \quad (3.19)$$

Moreover, there exists a constant $C_\Gamma > 0$, with the same dependence except for p , such that

$$\|\Gamma(t)\|_{L^\infty} \leq C_\Gamma e^{-\kappa t}, \quad 0 \leq t < T_*. \quad (3.20)$$

Proof. If $V_0 = 0$, (2.6) implies that u is spatially constant and $\Gamma \equiv 0$, so both conclusions are immediate. Suppose henceforth that $V_0 > 0$, and write

$$\phi = \Lambda^{-\alpha} u, \quad \Gamma = \nabla \phi.$$

The periodic convolution kernel H_α of $\Lambda^{-\alpha}$ is integrable, with a singularity of order $|h|^{\alpha-d}$, and has zero mean by (1.11). Hence

$$\phi(x, t) = \int_{\mathbb{T}^d} H_\alpha(x-y) (u(y, t) - u(x, t)) dy.$$

Using (2.6), we obtain

$$\|\phi(t)\|_{L^\infty} \leq \|H_\alpha\|_{L^1} V(t) \leq A_0 e^{-\kappa t}, \quad A_0 := \|H_\alpha\|_{L^1} V_0 > 0. \quad (3.21)$$

To proceed, we refine the estimate (3.5) as follows:

$$\begin{aligned} p|\Gamma|^{p-2}\Gamma \cdot \Lambda^\alpha \Gamma - \Lambda^\alpha(|\Gamma|^p) &\geq p|\Gamma|^{p-2}\Gamma \cdot \Lambda^\alpha \Gamma - \frac{p}{2}(|\Gamma|^2)^{\frac{p}{2}-1} \Lambda^\alpha(|\Gamma|^2) \\ &= p|\Gamma|^{p-2}\left(\Gamma \cdot \Lambda^\alpha \Gamma - \frac{1}{2}\Lambda^\alpha(|\Gamma|^2)\right) \geq c_p \frac{|\Gamma|^{p+\alpha}}{\|\phi\|_{L^\infty}^\alpha}, \end{aligned}$$

Here the first inequality is the Córdoba–Córdoba inequality for the convex function $s \mapsto s^{p/2}$ on $[0, \infty)$, with $s = |\Gamma|^2$. The last inequality follows from [6, Theorem 2.5], whose cutoff proof applies to periodic extensions using (1.12). If $\phi \equiv 0$, then $\Gamma \equiv 0$ and the quotient is understood as zero.

Inserting the refined estimate into (3.4), and using $\rho \geq \underline{\rho}$, (3.6), and (3.21), gives

$$\frac{d}{dt} \|\Gamma(t)\|_{L^p}^p + a_p e^{\alpha \kappa t} \|\Gamma(t)\|_{L^p}^{p+\alpha} \leq C_p G_* \|\Gamma(t)\|_{L^p}^p,$$

where $a_p := c_p \underline{\rho} A_0^{-\alpha} |\mathbb{T}^d|^{-\alpha/p} > 0$. Multiplying by $e^{p\kappa t}$ yields

$$\frac{d}{dt} \left(e^{\kappa t} \|\Gamma(t)\|_{L^p} \right)^p \leq -a_p \left(e^{\kappa t} \|\Gamma(t)\|_{L^p} \right)^{p+\alpha} + (C_p G_* + p\kappa) \left(e^{\kappa t} \|\Gamma(t)\|_{L^p} \right)^p.$$

Comparison with a constant supersolution gives

$$e^{\kappa t} \|\Gamma(t)\|_{L^p} \leq \max \left\{ \|\Gamma_0\|_{L^p}, \left(\frac{C_p G_* + p\kappa}{a_p} \right)^{1/\alpha} \right\},$$

which proves (3.19).

To obtain the endpoint estimate, fix once and for all $p > \max\{2, d/\alpha\}$, and put $\beta = d/(\alpha p) < 1$. By (3.6) and (3.19), $\|f(t)\|_{L^p} \leq C e^{-\kappa t}$. For $1 \leq t < T_*$, apply the componentwise Duhamel formula (3.18) on $[t-1, t]$ and the smoothing estimate (3.10) to obtain

$$\begin{aligned} \|\Gamma(t)\|_{L^\infty} &\leq C \|\Gamma(t-1)\|_{L^p} + C \int_{t-1}^t (1 + (t-s)^{-\beta}) \|f(s)\|_{L^p} ds \\ &\leq C e^{-\kappa(t-1)} + C e^{-\kappa t} \int_0^1 (1 + r^{-\beta}) e^{\kappa r} dr \leq C e^{-\kappa t}. \end{aligned}$$

For $0 \leq t < \min\{1, T_*\}$, applying (3.17) and increasing the constant by a factor e^κ gives

$$\|\Gamma(t)\|_{L^\infty} \leq C \left[1 + \left(t + \frac{t^{1-\beta}}{1-\beta} \right) e^{Ct} \right] \|\Gamma_0\|_{L^\infty} \leq C \left[1 + \frac{2-\beta}{1-\beta} e^C \right] \|\Gamma_0\|_{L^\infty} e^\kappa e^{-\kappa t}.$$

This proves (3.20) on the full existence interval. $\square$

4. PROPAGATION OF THE MODULI OF CONTINUITY

In this section, we propagate simultaneously a time-independent modulus of continuity for ρ and a time-dependent modulus of continuity for Γ . These quantities have invariant amplitude under the critical scaling (on periodic lifts, with the period rescaled)

$$\rho(x, t) \rightsquigarrow \rho(\lambda x, \lambda^\alpha t), \quad \Gamma(x, t) \rightsquigarrow \Gamma(\lambda x, \lambda^\alpha t).$$

The density bounds (2.4) and Theorem 3.5 give, for every $0 \leq t < T_*$,

$$\sup_{x, y \in \mathbb{T}^d} |\rho(x, t) - \rho(y, t)| \leq \bar{\rho}, \tag{4.1}$$

$$\sup_{x, y \in \mathbb{T}^d} |\Gamma(x, t) - \Gamma(y, t)| \leq M_\Gamma e^{-\kappa t}, \quad M_\Gamma := 2C_\Gamma. \tag{4.2}$$

Here $\kappa > 0$ is the rate in (2.6), and M_Γ is independent of the existence interval.

We follow the framework of [18, Section 3]: first choose the two moduli, describe their breakthrough scenarios, and then show that the time derivatives have the required signs at a putative contact.

The decay estimate (4.2) plays the role of the velocity-alignment estimate in that argument. The principal additional estimate concerns the commutator forcing in the equation for Γ . Throughout this section, constants denoted by C may depend on d, α , the initial data, and κ , but not on δ, λ , or time; a subscript records any additional dependence.

For points on $\mathbb{T}^d$, write $|x - y| := d_{\mathbb{T}^d}(x, y)$, and use shortest lifts when taking their difference in $\mathbb{R}^d$. All Euclidean integrals below act on periodic extensions.

4.1. The moduli of continuity and breakthrough scenarios. Recall the family of moduli used in [18, Section 3]:

$$\omega(\xi) = \omega_\lambda^\delta(\xi) := \begin{cases} \delta\lambda^{-1}\xi - \frac{1}{4}\delta\lambda^{-1-\frac{\alpha}{2}}\xi^{1+\frac{\alpha}{2}}, & 0 \leq \xi \leq \lambda, \\ \frac{3}{4}\delta + \frac{\delta}{2}\log \frac{\xi}{\lambda}, & \xi > \lambda, \end{cases} \quad (4.3)$$

where the parameters $0 < \delta < 1$ and $0 < \lambda < 1$ are to be determined. Note that ω is an increasing and concave function with $\omega(0) = 0$, $\omega'(0^+) = \delta\lambda^{-1}$, and $\omega''(0^+) = -\infty$. Define

$$\omega_1(\xi) := \omega(\xi), \quad \omega_2(\xi, t) := e^{-\kappa t}\omega(\xi). \quad (4.4)$$

Our goal is to show that, for all $0 \leq t < T_*$,

$$|\rho(x, t) - \rho(y, t)| < \omega_1(|x - y|), \quad (4.5)$$

and

$$|\Gamma(x, t) - \Gamma(y, t)| < \omega_2(|x - y|, t), \quad (4.6)$$

for every $x \neq y \in \mathbb{T}^d$. Periodic functions are identified with their periodic extensions when applying the modulus estimates. Note that ω_2 is chosen to be time-dependent in order to obtain the flocking estimate.

For any fixed $\delta > 0$, the initial data are Lipschitz, so choosing $\lambda > 0$ sufficiently small makes both ρ_0 and Γ_0 obey ω . We refer to [18, Lemma 3.1] for the standard fitting argument; its proof applies to the vector increment of Γ_0 without change. The propagated moduli imply

$$\|\nabla \rho(t)\|_{L^\infty} \leq \omega'_1(0^+) = \delta\lambda^{-1},$$

and

$$[\Gamma(t)]_{\text{Lip}} \leq \partial_\xi \omega_2(0^+, t) = \delta\lambda^{-1}e^{-\kappa t}.$$

Thus the second modulus also captures the decay of the potential gradient.

We use the standard breakthrough scenario of the modulus-of-continuity method [14, 18]. Suppose $t = t_1 \in (0, T_*)$ is the first time at which either (4.5) or (4.6) fails. Then there exist $x \neq y \in \mathbb{T}^d$, with

$$\xi = |x - y|,$$

such that either

$$\rho(x, t_1) - \rho(y, t_1) = \omega_1(\xi),$$

or

$$|\Gamma(x, t_1) - \Gamma(y, t_1)| = \omega_2(\xi, t_1). \quad (4.7)$$

Both modulus inequalities hold non-strictly on $[0, t_1]$ and strictly for $t < t_1$.

Accordingly, it suffices to prove, in the density scenario,

$$\partial_t(\rho(x, t) - \rho(y, t))|_{t=t_1} < 0, \quad (4.8)$$

and, in the Γ scenario,

$$\partial_t|\Gamma(x, t) - \Gamma(y, t)||_{t=t_1} + \kappa\omega_2(\xi, t_1) < 0. \quad (4.9)$$

The second condition is equivalent, at contact, to strict negativity of the derivative of $|\Gamma(x, t) - \Gamma(y, t)|/\omega_2(\xi, t)$, exactly as in [18, Section 3.1].

The amplitude bounds restrict the possible breakthrough distances. In the density scenario, (4.1) gives

$$0 < \xi \leq \Xi_1 := \omega^{-1}(\bar{\rho}) \leq \lambda \max \left\{ e^{2\bar{\rho}/\delta-3/2}, 1 \right\}.$$

In the Γ scenario, the common factor $e^{-\kappa t_1}$ in (4.2) and (4.7) cancels. Consequently,

$$0 < \xi \leq \Xi_2 := \omega^{-1}(M_\Gamma) \leq \lambda \max \left\{ e^{2M_\Gamma/\delta-3/2}, 1 \right\}.$$

Set

$$E := \max \left\{ e^{2\bar{\rho}/\delta-3/2}, e^{2M_\Gamma/\delta-3/2}, 1 \right\}. \quad (4.10)$$

Once δ is fixed, E is independent of λ and time. We shall choose λ so small that $E\lambda < 1/2$.

We next recall two standard dissipation bounds associated with the moduli (4.3). Suppose a function g obeys ω and $g(x) - g(y) = \omega(\xi)$. Then, we have

$$\Lambda^\alpha g(x) - \Lambda^\alpha g(y) \geq D_\alpha(\xi) > 0, \quad (4.11)$$

and

$$-\Lambda^\alpha g(x) \leq A_\alpha(\xi), \quad (4.12)$$

where

$$D_\alpha(\xi) := C_1 \left(\int_0^{\xi/2} \frac{2\omega(\xi) - \omega(\xi + 2\eta) - \omega(\xi - 2\eta)}{\eta^{1+\alpha}} d\eta + \int_{\xi/2}^\infty \frac{2\omega(\xi) - \omega(2\eta + \xi) + \omega(2\eta - \xi)}{\eta^{1+\alpha}} d\eta \right),$$

and

$$A_\alpha(\xi) := c_{d,\alpha} \text{p.v.} \int_{\mathbb{R}^d} \frac{\omega(|\xi e_1 - z|) - \omega(\xi)}{|z|^{d+\alpha}} dz.$$

With the special choice of ω in (4.3), [18, Lemmas 3.4 and 3.5] (with $\mu = \alpha/2$) give the following bounds

$$D_\alpha(\xi) \geq c_D \begin{cases} \delta \lambda^{-1-\frac{\alpha}{2}} \xi^{1-\frac{\alpha}{2}}, & 0 < \xi \leq \lambda, \\ \omega(\xi) \xi^{-\alpha}, & \xi > \lambda, \end{cases} \quad (4.13)$$

and

$$A_\alpha(\xi) \leq C \begin{cases} \delta \lambda^{-\frac{\alpha}{2}} \xi^{-\frac{\alpha}{2}}, & 0 < \xi \leq \lambda, \\ \delta \xi^{-\alpha}, & \xi > \lambda. \end{cases} \quad (4.14)$$

Next, since $\Gamma = \nabla \Lambda^{-\alpha}(u - \langle u \rangle)$ obeys ω_2 , we derive a modulus Ω for the velocity field u , using the relation

$$u - \langle u \rangle = - \sum_{j=1}^d \partial_j \Lambda^{\alpha-2} \Gamma_j.$$

Indeed, following [11, Equation (4.49)], we have

$$\Omega(\xi, t) = C \left(\int_0^\xi \frac{\omega_2(\eta, t)}{\eta^\alpha} d\eta + \xi \int_\xi^\infty \frac{\omega_2(\eta, t)}{\eta^{1+\alpha}} d\eta \right). \quad (4.15)$$

For the modulus (4.4), direct calculation gives

$$\Omega(\xi, t) \leq C e^{-\kappa t} \begin{cases} \delta \lambda^{-\alpha} \xi, & 0 < \xi \leq \lambda, \\ \omega(\xi) \xi^{1-\alpha}, & \xi > \lambda. \end{cases} \quad (4.16)$$

Letting $\xi \rightarrow 0$ in (4.16) gives

$$\|\nabla u(t)\|_{L^\infty} \leq C \delta \lambda^{-\alpha} e^{-\kappa t}, \quad 0 \leq t \leq t_1. \quad (4.17)$$

Finally, we record the control of the transported quantity $F = G/\rho$. Set

$$H := \frac{\partial_1 F}{\rho}.$$

Then, H satisfies the transport equation

$$(\partial_t + u\partial_1)H = 0.$$

Therefore

$$\|\partial_1 F(t)\|_{L^\infty} \leq \bar{\rho} \|H_0\|_{L^\infty}.$$

Differentiating (2.3) gives

$$(\partial_t + u\partial_1)\nabla F = -\nabla u \partial_1 F,$$

and hence, using (4.17),

$$\begin{aligned} \|\nabla F(t)\|_{L^\infty} &\leq \|\nabla F_0\|_{L^\infty} + \bar{\rho} \|H_0\|_{L^\infty} \int_0^t \|\nabla u(s)\|_{L^\infty} ds \\ &\leq \|\nabla F_0\|_{L^\infty} + C\kappa^{-1}\delta\lambda^{-\alpha}, \quad 0 \leq t \leq t_1. \end{aligned} \quad (4.18)$$

Since $G = F\rho$, the simultaneous modulus assumptions give

$$\begin{aligned} \|\nabla G(t)\|_{L^\infty} &\leq \bar{\rho} \|\nabla F(t)\|_{L^\infty} + \|F_0\|_{L^\infty} \|\nabla \rho(t)\|_{L^\infty} \\ &\leq C\lambda^{-1}, \quad 0 \leq t \leq t_1. \end{aligned} \quad (4.19)$$

In the last inequality we used $0 < \alpha < 1$ and $0 < \delta, \lambda < 1$. Both bounds are independent of the putative breakthrough time t_1 ; this is where the exponential factor in (4.17) is essential.

4.2. Evolution of the modulus of continuity of the density. The breakthrough estimate below follows the same argument as in [18]. In the breakthrough estimates below, omitted time arguments are understood to be t_1 , and time derivatives are evaluated at $t = t_1$, with x, y fixed. Starting from the dynamics

$$\partial_t \rho = -\rho \Lambda^\alpha \rho - F\rho^2 - u\partial_1 \rho,$$

we decompose

$$\begin{aligned} \partial_t \rho(x) - \partial_t \rho(y) &= -\rho(y)(\Lambda^\alpha \rho(x) - \Lambda^\alpha \rho(y)) - (\rho(x) - \rho(y))\partial_1 u(x) - \rho(y)F(x)(\rho(x) - \rho(y)) \\ &\quad - \rho(y)^2(F(x) - F(y)) - ((u\partial_1 \rho)(x) - (u\partial_1 \rho)(y)). \end{aligned}$$

Applying (4.11), (4.12), (4.18) and the velocity modulus (4.15), we obtain

$$\partial_t \rho(x, t_1) - \partial_t \rho(y, t_1) \leq -\underline{\rho} D_\alpha(\xi) + \omega(\xi)(A_\alpha(\xi) + 2\bar{\rho}\|F_0\|_{L^\infty}) + C(1 + \kappa^{-1}\delta\lambda^{-\alpha})\xi + \Omega(\xi)\omega'(\xi). \quad (4.20)$$

For $0 < \xi < \lambda$, we have $\omega(\xi) \leq \delta\xi/\lambda$ and $\omega'(\xi) \leq \delta/\lambda$. By (4.16), the transport term satisfies

$$\Omega(\xi)\omega'(\xi) \leq Ce^{-\kappa t_1}\delta^2\lambda^{-1-\alpha}\xi \leq C\delta(\delta\lambda^{-1-\alpha/2}\xi^{1-\alpha/2}).$$

Using (4.13) and (4.14) in (4.20), we obtain

$$\partial_t \rho(x, t_1) - \partial_t \rho(y, t_1) \leq \delta\lambda^{-1-\alpha/2}\xi^{1-\alpha/2}(-c_D\underline{\rho} + C\delta + C(\lambda^\alpha + \delta^{-1}\lambda^{1+\alpha} + \kappa^{-1}\lambda)). \quad (4.21)$$

For $\lambda < \xi \leq \Xi_1 \leq E\lambda$, we use $\omega'(\xi) = \delta/(2\xi)$ and $\omega(\xi) \geq 3\delta/4$. The transport term now satisfies

$$\Omega(\xi)\omega'(\xi) \leq Ce^{-\kappa t_1}\delta\omega(\xi)\xi^{-\alpha} \leq C\delta\omega(\xi)\xi^{-\alpha}.$$

The same bounds on D_α and A_α then give

$$\partial_t \rho(x, t_1) - \partial_t \rho(y, t_1) \leq \omega(\xi)\xi^{-\alpha}(-c_D\underline{\rho} + C\delta + C_E(\lambda^\alpha + \delta^{-1}\lambda^{1+\alpha} + \kappa^{-1}\lambda)). \quad (4.22)$$

Both (4.21) and (4.22) can be negative once δ is chosen sufficiently small and then λ is chosen sufficiently small. Thus the density breakthrough is ruled out.

4.3. Evolution of the modulus of continuity of Γ . Suppose that (4.7) holds at $t = t_1$. Set

$$\vartheta := \frac{\Gamma(x, t_1) - \Gamma(y, t_1)}{|\Gamma(x, t_1) - \Gamma(y, t_1)|}, \quad q(z, t) := \vartheta \cdot \Gamma(z, t).$$

We keep ϑ fixed and suppress the time variable t_1 below. Then q obeys the non-strict modulus ω_2 and

$$q(x) - q(y) = \omega_2(\xi).$$

The downward jump of ω_2' at λ excludes a contact at $\xi = \lambda$ for the smooth function q . Taking the difference of (3.2) at x and y and projecting onto ϑ , we obtain

$$\begin{aligned} \partial_t |\Gamma(x, t_1) - \Gamma(y, t_1)| &= -\rho(y)(\Lambda^\alpha q(x) - \Lambda^\alpha q(y)) - (\rho(x) - \rho(y))\Lambda^\alpha q(x) \\ &\quad - ((u\partial_1 q)(x) - (u\partial_1 q)(y)) + \vartheta \cdot (f(x) - f(y)). \end{aligned}$$

If $\rho(x) \geq \rho(y)$, we apply (4.11) and (4.12) to $e^{\kappa t_1} q$. If $\rho(x) < \rho(y)$, we instead write the first two terms as

$$-\rho(x)(\Lambda^\alpha q(x) - \Lambda^\alpha q(y)) + (\rho(y) - \rho(x))\Lambda^\alpha q(y),$$

and use $\Lambda^\alpha q(y) \leq e^{-\kappa t_1} A_\alpha(\xi)$, which follows by applying (4.12) to $-e^{\kappa t_1} q$ at the reversed contact. Thus, in both cases,

$$\partial_t |\Gamma(x, t_1) - \Gamma(y, t_1)| \leq e^{-\kappa t_1} \left(-\rho D_\alpha(\xi) + \omega(\xi) \max\{A_\alpha(\xi), 0\} + \Omega(\xi)\omega'(\xi) \right) + |f(x) - f(y)|. \quad (4.23)$$

The following proposition provides a key estimate on the commutator. The proof is postponed to Section 4.4.

Proposition 4.1. *Suppose that the simultaneous modulus inequalities hold non-strictly on $[0, t_1]$, with $0 < \delta, \lambda < 1$ and $E\lambda < 1/2$. For f in (3.2) and $\xi = |x - y|$, we have*

$$|f(x, t_1) - f(y, t_1)| \leq C e^{-\kappa t_1} \begin{cases} \delta \lambda^{-1} \xi, & 0 < \xi < \lambda, \\ \omega(\xi), & \lambda < \xi \leq E\lambda. \end{cases} \quad (4.24)$$

Here C depends only on the initial data, the structural parameters, and E , and is independent of λ and t_1 .

We now apply these estimates in the two regions.

For $0 < \xi < \lambda$, (4.16) gives

$$\Omega(\xi)\omega'(\xi) \leq C e^{-\kappa t_1} \delta^2 \lambda^{-1-\alpha} \xi \leq C \delta^2 \lambda^{-1-\alpha/2} \xi^{1-\alpha/2}. \quad (4.25)$$

The time derivative of the modulus contributes

$$\kappa \omega_2(\xi, t_1) \leq \kappa e^{-\kappa t_1} \delta \lambda^{-1} \xi \leq \kappa \lambda^\alpha e^{-\kappa t_1} \delta \lambda^{-1-\alpha/2} \xi^{1-\alpha/2}. \quad (4.26)$$

Applying (4.13), (4.14), (4.24), (4.25), and (4.26) to (4.23), we obtain

$$\partial_t |\Gamma(x, t_1) - \Gamma(y, t_1)| + \kappa \omega_2(\xi, t_1) \leq e^{-\kappa t_1} \delta \lambda^{-1-\alpha/2} \xi^{1-\alpha/2} \left(-c_D \rho + C\delta + C_E \lambda^\alpha + \kappa \lambda^\alpha \right). \quad (4.27)$$

For $\lambda < \xi \leq \Xi_2 \leq E\lambda$, we have

$$\Omega(\xi)\omega'(\xi) \leq C e^{-\kappa t_1} \delta \omega(\xi) \xi^{-\alpha} \leq C \delta \omega(\xi) \xi^{-\alpha}, \quad (4.28)$$

and

$$\kappa \omega_2(\xi, t_1) \leq E^\alpha \kappa \lambda^\alpha e^{-\kappa t_1} \omega(\xi) \xi^{-\alpha}. \quad (4.29)$$

Using (4.13), (4.14), (4.24), (4.28), and (4.29) in (4.23) gives

$$\partial_t |\Gamma(x, t_1) - \Gamma(y, t_1)| + \kappa \omega_2(\xi, t_1) \leq e^{-\kappa t_1} \omega(\xi) \xi^{-\alpha} \left(-c_D \rho + C\delta + C_E \lambda^\alpha + C_E \kappa \lambda^\alpha \right). \quad (4.30)$$

Both (4.27) and (4.30) are negative once δ is chosen sufficiently small and then λ is chosen sufficiently small. Thus (4.9) holds and the Γ breakthrough is ruled out.

4.4. The estimate on the commutator. Now, we focus on the estimate of the commutator term f , proving [Theorem 4.1](#). Recall the kernel representation

$$f_k(x) = -([\mathcal{R}_{k,\alpha}, u]G)(x) = - \int_{\mathbb{R}^d} J_k(z)(u(x-z) - u(x))G(x-z) dz, \quad (4.31)$$

where J_k is the Euclidean kernel of $\partial_k \Lambda^{-\alpha}$, with $|J_k(z)| \leq C|z|^{-d-1+\alpha}$ and $|\nabla J_k(z)| \leq C|z|^{-d-2+\alpha}$, and u, G are extended periodically to $\mathbb{R}^d$. This representation agrees with the periodic Fourier multiplier. The integral is absolutely convergent: smoothness controls the singularity at zero since $\alpha > 0$, and boundedness of the periodic functions controls infinity since $\alpha < 1$.

Fix $t = t_1$ and suppress the time variable. For $\xi = |x - y| \leq E\lambda < 1/2$, adding and subtracting a mixed term in [\(4.31\)](#) gives

$$\begin{aligned} f_k(x) - f_k(y) &= - \int_{\mathbb{R}^d} J_k(z) \left(u(x-z) - u(y-z) - u(x) + u(y) \right) G(y-z) dz \\ &\quad - \int_{\mathbb{R}^d} J_k(z) (u(x-z) - u(x)) (G(x-z) - G(y-z)) dz =: I_1 + I_2. \end{aligned}$$

For I_1 , we further write $G(y-z) = G(y) + (G(y-z) - G(y))$ and obtain

$$I_1 = -G(y)(\Gamma_k(x) - \Gamma_k(y)) - \int_{\mathbb{R}^d} J_k(z) \left(u(x-z) - u(y-z) - u(x) + u(y) \right) (G(y-z) - G(y)) dz.$$

Observe that

$$|u(x-z) - u(y-z) - u(x) + u(y)| \leq 2 \min\{\Omega(\xi), \Omega(|z|)\}.$$

Moreover, by [\(4.19\)](#) and $\|G\|_{L^\infty} \leq G_*$, we have

$$|G(y-z) - G(y)| \leq C \min\{|z|/\lambda, 1\}.$$

Taking absolute values and integrating radially, we obtain

$$|I_1| \leq G_* \omega_2(\xi, t_1) + C \int_0^\infty \min\{r/\lambda, 1\} \min\{\Omega(r), \Omega(\xi)\} r^{\alpha-2} dr. \quad (4.32)$$

For $0 < \xi < \lambda$, splitting the integral in [\(4.32\)](#) at ξ and λ , and using [\(4.16\)](#) gives

$$\begin{aligned} |I_1| &\leq C e^{-\kappa t_1} \delta \lambda^{-1} \xi + C \left(\lambda^{-1} \int_0^\xi \Omega(r) r^{\alpha-1} dr + \lambda^{-1} \Omega(\xi) \int_\xi^\lambda r^{\alpha-1} dr + \Omega(\xi) \int_\lambda^\infty r^{\alpha-2} dr \right) \\ &\leq C e^{-\kappa t_1} \delta \left(\lambda^{-1} \xi + \lambda^{-1-\alpha} \xi^{1+\alpha} \right) \leq C e^{-\kappa t_1} \delta \lambda^{-1} \xi. \end{aligned}$$

For $\lambda < \xi \leq E\lambda$, we instead split [\(4.32\)](#) at λ and ξ :

$$\begin{aligned} |I_1| &\leq G_* \omega_2(\xi, t_1) + C \left(\lambda^{-1} \int_0^\lambda \Omega(r) r^{\alpha-1} dr + \int_\lambda^\xi \Omega(r) r^{\alpha-2} dr + \Omega(\xi) \int_\xi^\infty r^{\alpha-2} dr \right) \\ &\leq C e^{-\kappa t_1} \left(\omega(\xi) + \delta + \int_\lambda^\xi \frac{\omega(r)}{r} dr \right) \leq C_E e^{-\kappa t_1} \omega(\xi). \end{aligned}$$

Here we have used $\omega(\xi) \geq 3\delta/4$ and $\xi/\lambda \leq E$.

For I_2 , we split the integral at $|z| = \lambda$. In the near field, [\(4.19\)](#) and the velocity modulus give

$$\left| \int_{|z| \leq \lambda} J_k(z) (u(x-z) - u(x)) (G(x-z) - G(y-z)) dz \right| \leq C \lambda^{-1} \xi \int_0^\lambda \Omega(r) r^{\alpha-2} dr.$$

In the far field, we transfer the difference of G onto the kernel and the velocity increment. Choose shortest lifts of x, y . Then

$$G(x-z) - G(y-z) = - \int_0^1 (x-y) \cdot \nabla_z G(y + s(x-y) - z) ds.$$

Integrating by parts on $\{|z| > \lambda\}$ yields

$$\begin{aligned} & \left| \int_{|z|>\lambda} J_k(z) (u(x-z) - u(x)) (G(x-z) - G(y-z)) dz \right| \\ & \leq G_* \xi \left[\int_{|z|=\lambda} |J_k(z)| |u(x-z) - u(x)| dS(z) + \int_{|z|>\lambda} |\nabla_z (J_k(z) (u(x-z) - u(x)))| dz \right] \\ & \leq CG_* \xi \|\nabla u\|_{L^\infty} \left(\lambda^{\alpha-1} + \int_\lambda^\infty r^{\alpha-2} dr \right) \leq C\xi \lambda^{\alpha-1} \|\nabla u\|_{L^\infty}. \end{aligned}$$

Here we used $|u(x-z) - u(x)| \leq \|\nabla u\|_{L^\infty} |z|$. The boundary term at infinity vanishes, since the velocity increment is bounded by $V(t_1)$ and $\alpha < 1$. Combining the two regions gives

$$|I_2| \leq C\lambda^{-1}\xi \int_0^\lambda \Omega(r) r^{\alpha-2} dr + C\xi \lambda^{\alpha-1} \|\nabla u\|_{L^\infty}. \quad (4.33)$$

For $0 < \xi < \lambda$, (4.16), (4.17), and (4.33) give

$$|I_2| \leq Ce^{-\kappa t_1} \delta \xi \left(\lambda^{-1-\alpha} \int_0^\lambda r^{\alpha-1} dr + \lambda^{\alpha-1} \lambda^{-\alpha} \right) \leq Ce^{-\kappa t_1} \delta \lambda^{-1} \xi. \quad (4.34)$$

For $\lambda < \xi \leq E\lambda$, the estimate (4.34) is valid for every ξ and gives

$$|I_2| \leq Ce^{-\kappa t_1} \delta \frac{\xi}{\lambda} \leq C_E e^{-\kappa t_1} \omega(\xi).$$

Combining all the bounds proves (4.24), and hence Theorem 4.1.

4.5. Closing the simultaneous modulus argument. We choose the parameters in the same order as in [18, Section 3]. First fix $\delta > 0$ sufficiently small that all the $C\delta$ terms in (4.21), (4.22), (4.27), and (4.30) satisfy

$$C\delta \leq \frac{1}{4} c_D \rho_-.$$

Then E in (4.10) is fixed. Choose $\lambda > 0$ sufficiently small that

- (i) the initial data ρ_0 and Γ_0 obey ω_λ^δ ;
- (ii) $E\lambda < 1/2$;
- (iii) the lower-order density terms satisfy

$$C_E \left(\lambda^\alpha + \delta^{-1} \lambda^{1+\alpha} + \kappa^{-1} \lambda \right) \leq \frac{1}{4} c_D \rho_-;$$

- (iv) the commutator and the derivative of the time-dependent modulus satisfy

$$C_E \lambda^\alpha (1 + \kappa) \leq \frac{1}{4} c_D \rho_-.$$

Such a choice is possible since $\alpha > 0$, with δ and E already fixed. All choices depend only on the initial data and the structural parameters, and are independent of t_1 or any finite time horizon.

With these choices, (4.21) and (4.22) imply (4.8), while (4.27) and (4.30) imply (4.9). Neither breakthrough can therefore occur. We have proved, for every $0 \leq t < T_*$ and $x \neq y$,

$$|\rho(x, t) - \rho(y, t)| < \omega_1(|x - y|), \quad |\Gamma(x, t) - \Gamma(y, t)| < \omega_2(|x - y|, t).$$

In particular,

$$\sup_{0 \leq t < T_*} \|\nabla \rho(t)\|_{L^\infty} \leq \delta \lambda^{-1}, \quad (4.35)$$

$$[\Gamma(t)]_{\text{Lip}} \leq \delta \lambda^{-1} e^{-\kappa t}, \quad 0 \leq t < T_*, \quad (4.36)$$

and (4.17) yields

$$\|\nabla u(t)\|_{L^\infty} \leq C \delta \lambda^{-\alpha} e^{-\kappa t}, \quad 0 \leq t < T_*. \quad (4.37)$$

Since conservation of mass and momentum gives $\|u(t) - \bar{u}\|_{L^\infty} \leq V(t)$, (2.6) also gives exponential decay of $u - \bar{u}$ in $W^{1,\infty}$, at the same rate κ . The uniform bounds also close the continuation argument in Section 5.

5. GLOBAL REGULARITY AND ASYMPTOTIC ALIGNMENT

We complete the proof of Theorem 1.1 by verifying the Beale–Kato–Majda-type continuation criterion in Theorem 2.1; see [16, Theorem 1.1]. We then prove exponential convergence of the density to a traveling profile.

Proof of Theorem 1.1. Let (ρ, u) be the maximal classical solution of Theorem 2.1, with existence time T_* . Suppose, for contradiction, that $T_* < \infty$. The coefficient bounds (2.4) and (2.5) hold on $[0, T_*)$, and Theorem 3.5 gives

$$\|\Gamma(t)\|_{L^\infty} \leq C_\Gamma e^{-\kappa t}, \quad 0 \leq t < T_*.$$

Consequently the parameter choice in Section 4.5 can be made once on the full existence interval: first fix δ , then E , and finally a positive λ . These choices depend only on the initial data and the structural bounds, and are independent of a putative breakthrough time $t_1 < T_*$. The argument of Section 4 thus gives

$$\sup_{0 \leq t < T_*} (\|\nabla \rho(t)\|_{L^\infty} + [\Gamma(t)]_{\text{Lip}}) \leq 2\delta\lambda^{-1} < \infty.$$

Combining this with (4.37) yields

$$\sup_{0 \leq t < T_*} (\|\nabla \rho(t)\|_{L^\infty} + \|\nabla u(t)\|_{L^\infty}) < \infty.$$

This contradicts (2.2). Hence $T_* = \infty$.

Existence, uniqueness, and the Sobolev regularity (1.7) now follow by continuation of the local solution. The uniform positive lower and upper density bounds are given by (2.4). The parameters in Section 4.5 are independent of time, so (4.35) gives a uniform Lipschitz bound for ρ , while (4.36) and (4.37) give

$$[\Gamma(t)]_{\text{Lip}} + \|\nabla u(t)\|_{L^\infty} \leq C e^{-\kappa t}, \quad t \geq 0.$$

Combining the velocity-gradient bound with $\|u(t) - \bar{u}\|_{L^\infty} \leq V(t)$ and (2.6) proves (1.9).

For the density alignment, we follow Lear and Shvydkoy [16, Step 4.4, p. 827]. Define the density in the moving frame by

$$\tilde{\rho}(x, t) := \rho(x + \bar{u}te_1, t).$$

The continuity equation gives

$$\partial_t \tilde{\rho}(x, t) = -[(u - \bar{u})\partial_1 \rho + \rho\partial_1 u](x + \bar{u}te_1, t).$$

Using (2.4), (4.35), and (1.9), we obtain

$$\|\partial_t \tilde{\rho}(t)\|_{L^\infty} \leq \|u(t) - \bar{u}\|_{L^\infty} \|\nabla \rho(t)\|_{L^\infty} + \|\rho(t)\|_{L^\infty} \|\nabla u(t)\|_{L^\infty} \leq C e^{-\kappa t}.$$

Hence, for $s \geq t \geq 0$,

$$\|\tilde{\rho}(s) - \tilde{\rho}(t)\|_{L^\infty} \leq C \int_t^s e^{-\kappa r} dr \leq \frac{C}{\kappa} e^{-\kappa t}.$$

Thus $\tilde{\rho}(t)$ converges uniformly to a profile ρ_∞ . The uniform Lipschitz bound (4.35) passes to the limit, so $\rho_\infty \in W^{1,\infty}(\mathbb{T}^d)$. The density bounds and conservation of mass also pass to the limit. Letting $s \rightarrow \infty$ in the preceding estimate and translating back to the original frame proves (1.8), with the same rate κ . $\square$

APPENDIX A. THE PERIODIC ENDPOINT COMMUTATOR ESTIMATE

We give a short localization argument to deduce [Theorem 2.2](#) from the Euclidean commutator estimate.

Proof of Theorem 2.2. Set $s = 1 - \alpha \in (0, 1)$ and fix $1 < p < \infty$. Write $A = \mathcal{R}_{k,\alpha}$ and let A_E be its Euclidean counterpart, with symbol $i\xi_k|\xi|^{-\alpha}$. The Euclidean estimate (see, e.g., [17, Corollary 1.4(2)]) gives

$$\| [A_E, \tilde{v}] \tilde{h} \|_{L^p(\mathbb{R}^d)} \leq C \| \Lambda_E^s \tilde{v} \|_{L^p(\mathbb{R}^d)} \| \tilde{h} \|_{L^\infty(\mathbb{R}^d)}, \quad \tilde{v}, \tilde{h} \in C_c^\infty(\mathbb{R}^d), \quad (\text{A.1})$$

where $\Lambda_E^s = (-\Delta_{\mathbb{R}^d})^{s/2}$.

Let v, h be smooth periodic functions, identified with their periodic extensions. Choose $\chi \in C_c^\infty(\mathbb{R}^d)$ with $0 \leq \chi \leq 1$, $\chi = 1$ on $[-1, 1]^d$, and support in $[-2, 2]^d$. Set $\chi_L(x) = \chi(x/L)$ and $Q_L = [-L, L]^d$, for $L \geq 1$. For $B = A_E$ or Λ_E^s , the difference kernel satisfies $|K_B(z)| \leq C|z|^{-d-s}$. Hence, for every smooth periodic g , Minkowski's inequality gives

$$\begin{aligned} \| [B, \chi_L] g \|_{L^p(\mathbb{R}^d)} &= \| B(\chi_L g) - \chi_L Bg \|_{L^p(\mathbb{R}^d)} \leq C \| g \|_{L^\infty} \int_{\mathbb{R}^d} \frac{\| \chi_L - \chi_L(\cdot - z) \|_{L^p(\mathbb{R}^d)}}{|z|^{d+s}} dz \\ &\leq C \| g \|_{L^\infty} L^{d/p} \int_0^\infty \min\{r/L, 1\} \frac{dr}{r^{1+s}} \leq C \| g \|_{L^\infty} L^{d/p-s}. \end{aligned} \quad (\text{A.2})$$

Here Bg , for periodic g , is defined through the absolutely convergent difference integral and agrees with the corresponding periodic Fourier multiplier. The same estimate holds with χ_L replaced by χ_L^2 .

On $\mathbb{R}^d$, we have the identity

$$\chi_L^2 [A, v] h = [A_E, \chi_L v] (\chi_L h) - [A_E, \chi_L^2] (vh) + \chi_L v [A_E, \chi_L] h.$$

Since $\chi_L = 1$ on Q_L and $0 \leq \chi_L \leq 1$, the triangle inequality and (A.2) with $B = A_E$ give

$$\begin{aligned} \| [A, v] h \|_{L^p(Q_L)} &\leq \| [A_E, \chi_L v] (\chi_L h) \|_{L^p(\mathbb{R}^d)} + \| [A_E, \chi_L^2] (vh) \|_{L^p(\mathbb{R}^d)} + \| v \|_{L^\infty} \| [A_E, \chi_L] h \|_{L^p(\mathbb{R}^d)} \\ &\leq \| [A_E, \chi_L v] (\chi_L h) \|_{L^p(\mathbb{R}^d)} + C \| v \|_{L^\infty} \| h \|_{L^\infty} L^{d/p-s}. \end{aligned}$$

For the first commutator, applying (A.1) with $\tilde{v} = \chi_L v$ and $\tilde{h} = \chi_L h$ yields

$$\| [A_E, \chi_L v] (\chi_L h) \|_{L^p(\mathbb{R}^d)} \leq C \| \Lambda_E^s (\chi_L v) \|_{L^p(\mathbb{R}^d)} \| h \|_{L^\infty}.$$

Using (A.2) with $B = \Lambda_E^s$, we further obtain

$$\| \Lambda_E^s (\chi_L v) \|_{L^p(\mathbb{R}^d)} \leq \| \chi_L \Lambda_E^s v \|_{L^p(\mathbb{R}^d)} + \| [\Lambda_E^s, \chi_L] v \|_{L^p(\mathbb{R}^d)} \leq CL^{d/p} \| \Lambda^s v \|_{L^p(\mathbb{T}^d)} + C \| v \|_{L^\infty} L^{d/p-s},$$

where we used the periodicity of $\Lambda^s v$ and the support property $\text{supp } \chi_L \subset [-2L, 2L]^d$.

Combining the preceding estimates gives

$$\| [A, v] h \|_{L^p(Q_L)} \leq CL^{d/p} \| \Lambda^s v \|_{L^p(\mathbb{T}^d)} \| h \|_{L^\infty(\mathbb{T}^d)} + C \| v \|_{L^\infty(\mathbb{T}^d)} \| h \|_{L^\infty(\mathbb{T}^d)} L^{d/p-s}.$$

Now take $L = N\pi$, $N \in \mathbb{N}$. Since $Q_{N\pi}$ is the union of N^d fundamental 2π -periodic cells,

$$\| [A, v] h \|_{L^p(Q_{N\pi})} = N^{d/p} \| [A, v] h \|_{L^p(\mathbb{T}^d)}.$$

Dividing the preceding estimate by $N^{d/p}$ and absorbing the fixed powers of π into the constant, we obtain

$$\| [A, v] h \|_{L^p(\mathbb{T}^d)} \leq C \| \Lambda^s v \|_{L^p(\mathbb{T}^d)} \| h \|_{L^\infty(\mathbb{T}^d)} + CN^{-s} \| v \|_{L^\infty(\mathbb{T}^d)} \| h \|_{L^\infty(\mathbb{T}^d)}.$$

Letting $N \rightarrow \infty$ and using $s = 1 - \alpha > 0$, we conclude that

$$\| [\mathcal{R}_{k,\alpha}, v] h \|_{L^p(\mathbb{T}^d)} \leq C \| \Lambda^{1-\alpha} v \|_{L^p(\mathbb{T}^d)} \| h \|_{L^\infty(\mathbb{T}^d)}.$$

Finally, the L^p -boundedness of the periodic Riesz transforms implies

$$\left\| \Lambda^{1-\alpha} v \right\|_{L^p} \leq C \left\| \nabla \Lambda^{-\alpha} v \right\|_{L^p}.$$

This proves (2.7) for smooth periodic functions. The general case $v, h \in C^1(\mathbb{T}^d)$ follows by periodic mollification. $\square$

Declaration on the use of generative AI. During the preparation of this manuscript, the authors used ChatGPT (OpenAI) for language refinement, LaTeX formatting, and as an auxiliary tool to review logical consistency across proof steps. All mathematical content and arguments were independently analyzed, reconstructed, and verified by the authors, who take full responsibility for the accuracy and originality of the published work.

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(Changhui Tan)

DEPARTMENT OF MATHEMATICS, UNIVERSITY OF SOUTH CAROLINA, COLUMBIA SC 29208, USA

Email address: `tan@math.sc.edu`

(Liutang Xue)

SCHOOL OF MATHEMATICAL SCIENCES, LABORATORY OF MATHEMATICS AND COMPLEX SYSTEMS (MOE), BEIJING NORMAL UNIVERSITY, BEIJING 100875, P.R. CHINA

Email address: `xueli@bnu.edu.cn`